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BATCH 25

SCHOOLOFCOMPUTERSC IENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENTOFCOMPUTERSCIENCE ENGINEERING	
ProgramName:B.Tech		AssignmentType:Lab	AcademicYear: 2025-2026
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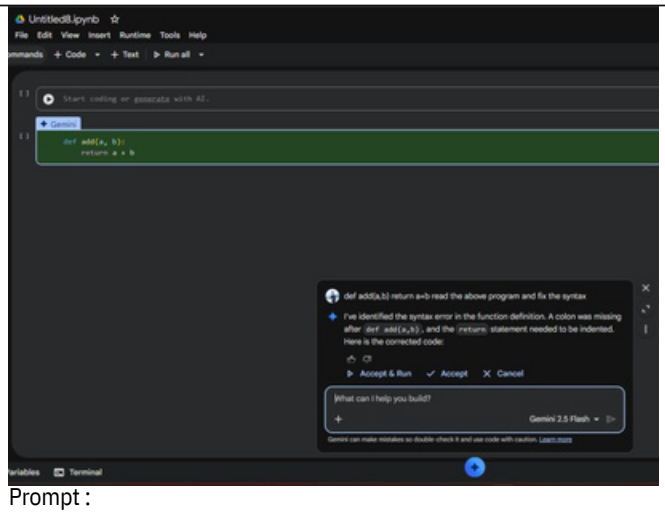
Course Code	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week4 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
AssignmentNumber:7.3(Present assignment number)/24(Total number of assignments)			

	Question	Expected Time to complete
	Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs Lab Objectives	
	• To identify and correct syntax, logic, and runtime errors in Python programs using AI tools • To understand common programming bugs and AI-assisted debugging suggestions • To evaluate how AI explains, detects, and fixes different types of coding errors • To build confidence in using AI for structured debugging practices	
	Lab Outcomes (LOs) After completing this lab, students will be able to: • Use AI tools to detect and correct syntax, logic, and runtime errors • Interpret AI-suggested bug fixes and explanations • Apply systematic debugging strategies using AI-generated insights • Refactor buggy code using reliable programming patterns	
	Task 1: Fixing Syntax Errors	

1	Scenario
	YouarereviewingaPythonprogramwhereabasicfunctiondefinitioncontainsa syntax error.
	python
	`def add(a, b)`
	`return a + b`
	Requirements
	• Provide a Python function add(a, b) with a missing colon
	• Use an AI tool to detect the syntax error
	• Allow AI to correct the function definition
	• Observe how AI explains the syntax issue
	Expected Output
	• Corrected function with proper syntax
	• Syntax error resolved successfully
	• AI-generated explanation of the fix
		W e ek4 - Wednesday



Prompt :

def add(a,b) return a+b read the above program and fix the syntax

I've identified the syntax error in the function definition. A colon was missing after def add(a,b), and the return statement needed to be indented. Here is the corrected code:

Task 2: Debugging Logic Errors in Loops

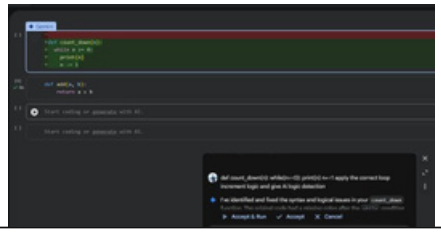
Scenario

You are debugging a loop that runs infinitely due to a logical mistake.

```python
def count_down(n):
    while n >= 0:
        print(n)
        n += 1  # Should be n -= 1
```

Requirements
- Provide a loop with an increment or decrement error
- Use AI to identify the cause of infinite iteration
- Let AI fix the loop logic
- Analyze the corrected loop behavior

Expected Output
- Infinite loop issue resolved
- Correct increment/decrement logic applied
- AI explanation of the logic error



Prompt

def count_down(n): while(n>=0): print(n) n+=1 apply the correct loop increment logic and give Ai logic detection

I've identified and fixed the syntax and logical issues in your count_down function. The original code had a missing colon after the while condition and incorrect indentation. More critically, for a countdown function, the variable n needs to be decremented (n -= 1) within the loop, not incremented (n += 1), otherwise, it would result in an infinite loop if n starts as a non-negative number.

Task 3: Handling Runtime Errors (Division by Zero)

Scenario

A Python function crashes during execution due to a division by zero error.

```
# Debug the following code
def divide(a, b):
    return a / b

print(divide(10, 0))
```

Requirements
• Provide a function that performs division without validation
• Use AI to identify the runtime error
• Let AI add try-except blocks for safe execution
• Review AI's error-handling approach
Expected Output

• Function executes safely without crashing • Division by zero handled using try-except • Clear AI-generated explanation of runtime error handling

Prompt: def divide(a,b): return a/b print(divide(10,0)) debug the above code

The code you provided attempts to divide by zero, which causes a ZeroDivisionError in Python. To handle this, I've modified the divide function to use a try-except block. This allows the program to catch the ZeroDivisionError and return an informative message instead of crashing. The code in cell 31be9d13 has executed successfully. As intended, the divide function correctly handled the division by zero, printing "Error: Cannot divide by zero!", and then successfully calculated 10 divided by 2,

which resulted in 5.0.

Task 4: Debugging Class Definition Errors

Scenario

You are given a faulty Python class where the constructor is incorrectly defined.

```python
class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```

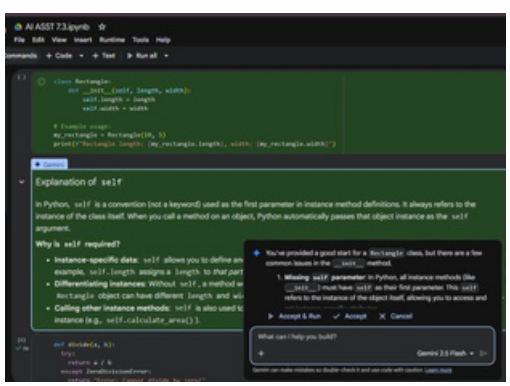
Requirements

- Provide a class definition with missing self-parameter
- Use AI to identify the issue in the init __() method
- Allow AI to correct the class definition
- Understand why self is required

Expected Output

- Corrected init ()method
- Proper use of self in class definition
- AI explanation of object-oriented error

python class Rectangle: def _init_(length, width): self.length = length self.width = width Requirements · Provide a class definition with missing self-parameter



In Python, self is a convention (not a keyword) used as the first parameter in instance method definitions. It always refers to the instance of the class itself. When you call a method on an object, Python automatically passes that object instance as the self argument.

Why is self required?

- Instance-specificdata : self allows you to define and access attributes (data) that belong to a specific instance of the class. For example, self.length assigns a length to that particular Rectangle object.
- Differentiating instances: Without self, a method wouldn't know which object's attributes it should be operating on. Each Rectangle object can have different length and width values, and self ensures you're working with the correct object's data.
- Calling other instance methods: self is also used to call other methods defined within the same class on the current object instance (e.g., `self.calculate_area()`).

Task 5: Resolving Index Errors in Lists

Scenario

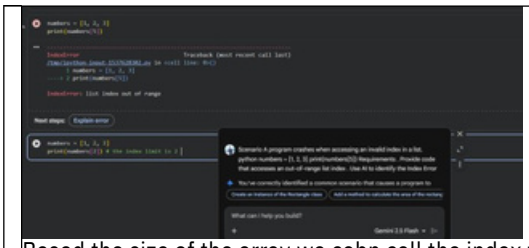
A program crashes when accessing an invalid index in a list.

```python
numbers = [1, 2, 3]
print(numbers[5])
```

Requirements
- Provide code that accesses an out-of-range list index
- Use AI to identify the Index Error
- Let AI suggest safe access methods
- Apply bounds checking or exception handling

Expected Output
- Index error resolved
- Safe list access logic implemented
- AI suggestion using length checks or exception handling

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots



Based the size of the array we cabn call the index to print

We must call it on its indexs ranges .